

Competition Day Policy Document

Synchrophasor Anomaly Detection Competition
Test-Day Instructions, Submission Format & Deadlines

Greetings, and welcome to competition day!

This document outlines the instructions and the required format for testing your algorithm and uploading your results on the day of the SGSMA 2026 Synchrophasor Anomaly Detection Competition. Please read it carefully before the test data is released. Following the format exactly ensures your submission can be evaluated fairly and on time.

At a Glance

-  You will receive **two test datasets** to evaluate your algorithm.
-  Test data is released at **9:50 AM**; all submissions are due by **12:00 PM**.
-  The `label` column is left **blank** — your job is to fill it in.
-  Submit **one folder** named after you, containing the files described in Section 5.

1 Competition Day Schedule

The test phase follows the timeline below. All times are local to the conference venue. Please plan to have your training pipeline ready *before* the test data is released so that you can devote the full window to inference and packaging your results.

Table 1: Competition day timeline.

Time	Activity
9:50 AM	Two test datasets released on the shared drive under <code>PMU_Data_Testing</code> .
9:50 AM–12:00 PM	Participants run their algorithms on both test datasets and prepare their submission folder.
12:00 PM	Hard deadline. All submissions must be uploaded to the submission drive. Late submissions will not be accepted.

2 Accessing the Test Data

The test data will be available on the **same drive** where the training data was hosted, in a folder named `PMU_Data_Testing`, at approximately **9:50 AM** on the day of the competition.

Where to download the test data

Use the link provided on the SGSMA competition website to view and download the test data:

[A Dropbox folder will be shared with you!](#)

Confirm your access early

Make sure you can reach the shared drive and the submission link *before* 9:50 AM. If you cannot access either, contact the coordinator (Section 7) as soon as possible. Do not wait until the deadline to discover an access problem.

3 The Two Test Datasets

On competition day you will be given **two separate test datasets**. Each dataset contains approximately **25 minutes** of PMU measurements, in the *exact same format* as the training data, with one key difference: the `label` column is left **blank**.

What to expect in each test set

- **Same column structure** as the training data (timestamps, three-phase voltage and current phasors, frequency, ROCOF, and the `DATA_PRESENT` flag).
- **Same 8 PMU buses** from the IEEE 39-bus system, with timestamps aligned across all bus files.
- **Approximately 25 minutes** of data per set (~45,000 rows per PMU at 30 fps).
- **The `label` column is empty** — it is the only column you are required to fill in.

ⓘ Both test cases are mandatory

You must run your algorithm on **both** test datasets and submit labeled results for **each**. A submission that covers only one of the two test cases is considered incomplete.

4 What Your Algorithm Must Produce

For every time step (row) in each test dataset, your algorithm should generate the labels corresponding to the three competition tasks, consistent with the training-phase event label definitions (labels 0–8):

1. **Detection** — is the system in normal operation or in an abnormal condition?
2. **Classification** — if abnormal, which event type is occurring?
3. **Localization** — where in the network did the event originate (bus and/or line)?

You generate these labels for the data points and submit them by **appending them to the last column** (the blank `label` column) of each test dataset.

5 Submission Folder Structure

Your entire submission must be packaged in a **single folder** named after your first initial and last name, followed by the competition tag. For example, if your name is **John Smith**, the folder must be named:

J_Smith_SGSMA2026

That folder must contain the files described below.

📁 Required contents of J_Smith_SGSMA2026/

1. **Results File for Test Case 1 (XLSX)** — J_Smith_Results_Test1.xlsx
2. **Results File for Test Case 2 (XLSX)** — J_Smith_Results_Test2.xlsx

Each results file must also report your model's **accuracy** and **F1 score** for that test case (see Section 5.2).

5.1 Results Files (XLSX) — One per Test Case

For each of the two test datasets, submit a results file in `.xlsx` format. Each file must be the provided test dataset with your predicted labels **appended to the blank `label` column**, leaving every other column unchanged.

📁 Naming the results files

Name each results file using your first initial and last name plus the test-case number, e.g.:

- J_Smith_Results_Test1.xlsx
- J_Smith_Results_Test2.xlsx

⚠ Preserve the original format

Do **not** reorder rows, rename columns, or change timestamp formatting. Fill in *only* the blank `label` column. Misaligned timestamps or altered columns may make your results impossible to score.

5.2 Performance Metrics to Report

In addition to the labeled results, each participant must report the performance of their model **for each of the two test cases separately**. Specifically, report:

- **Accuracy** of your model on Test Case 1 and on Test Case 2.
- **F1 score** of your model on Test Case 1 and on Test Case 2.

i Where to report your metrics

Include these values on a separate sheet named **Metrics** inside each results workbook (or clearly labeled at the top of the results sheet), so that each test case reports its own accuracy and F1 score. A suggested layout is shown below.

Table 2: Performance metrics to report for each test case.

Metric	Test Case 1	Test Case 2
Accuracy	(your value)	(your value)
F1 Score	(your value)	(your value)

6 Submission Procedure and Deadline

All submissions must be uploaded to the submission drive link provided on the SGSMA competition website **before the competition ends at 12:00 PM**.

🕒 Final deadline — 12:00 PM

The submission drive closes at **12:00 PM** sharp. Upload your complete folder well before the deadline; do not wait until the final minutes. Submissions received after 12:00 PM cannot be accepted.

💡 Pre-submission checklist

- ✓ Folder named `<Initial>_<LastName>_SGSMA2026`.
- ✓ Results file for **Test Case 1** (XLSX) with labels appended.
- ✓ Results file for **Test Case 2** (XLSX) with labels appended.
- ✓ Only the blank `label` column was modified in each results file.
- ✓ Accuracy and F1 score reported for **each** test case.
- ✓ Folder uploaded to the submission drive before 12:00 PM.

7 Questions and Contact

If you have any questions or require clarification regarding the competition instructions, dataset, evaluation criteria, or any other related matters, please contact the coordinator:

Competition Coordinator

Masoud Barati

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Good Luck on Competition Day!

Smart Grid Synchrophasor Measurement & Analytics — SGSMA 2026